Lunar Surface Analysis via XRF

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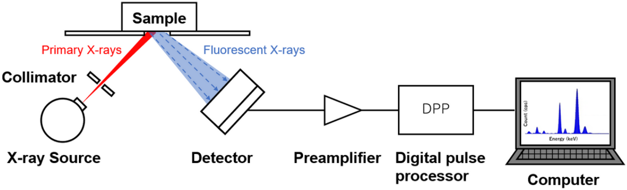
Introduction

- **Goal:** Identification and quantification of chemical elements on the lunar surface.
- **Challenge:** Remote sensing via X-ray Fluorescence (XRF) produces complex, noisy, and high-dimensional spectral data.
- **Methodology:** Using Machine Learning to automate and refine the interpretation of these spectra.
- **Data Strategy:** Development of a custom dataset and utilization of `multiel-spectra` for high-fidelity synthetic data generation.

XRF Spectroscopy Fundamentals

- High-energy X-rays/Gamma rays irradiate the sample.
- Secondary X-rays (fluorescence) are emitted, unique to each element.
- The detector captures a spectrum where energy peaks correspond to specific atoms.



Project Phases

The analysis focuses on determining the chemical elements within the XRF spectra through three interconnected computational tasks:

1. Classification
2. Regression
3. Denoising

Synthetic Data Generation

To train models, we developed a Generator that simulates physical variations in XRF spectra.

- **Composition:** Samples contain **0 to 5** elements randomly selected from **41** possible candidates.
- **Stochasticity:** To ensure model robustness, each spectrum is generated by randomly varying operating conditions and noise levels simulating real-world measurement variability.

Classification

The classification task focus on identifying the elemental composition by analyzing spectral signatures.

Analytical stages

- **Multi-label classification** across a pool of 41 potential elements.
- **Multi-class classification** across 5 classes (1 to 5 elements) defined by the number of elements present in the sample.

Preprocessing Strategies and Architectures

Preprocessing Scaling

Tested to handle high dynamic range in spectral counts:

- **Standard:** 0 mean and 1 variance.
- **MinMax:** Rescaling to $[0, 1]$ range.
- **Log-MinMax:** Logarithmic transformation to compress peaks.

Model Architectures

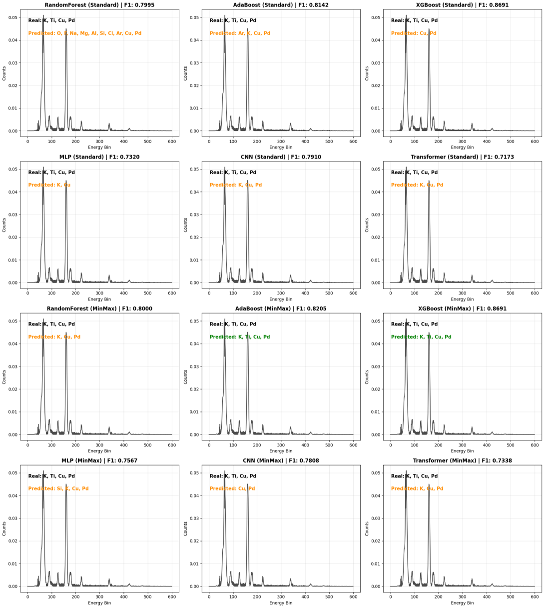
Machine Learning:

- Random Forest, AdaBoost, XGBoost

Deep Learning:

- MLP, CNN, Transformer

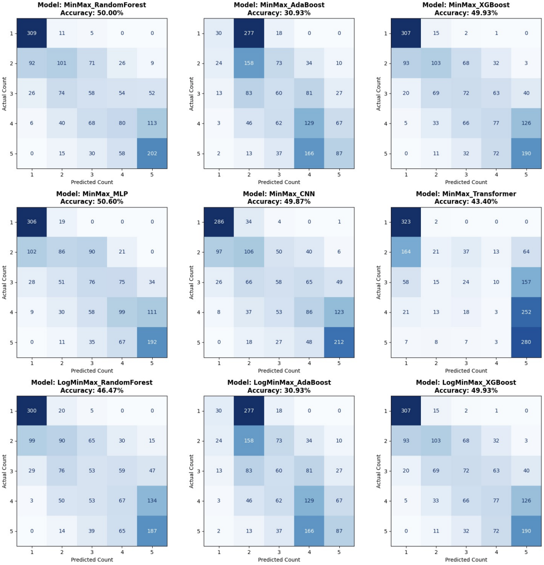
Results: Example Multi-label Classification



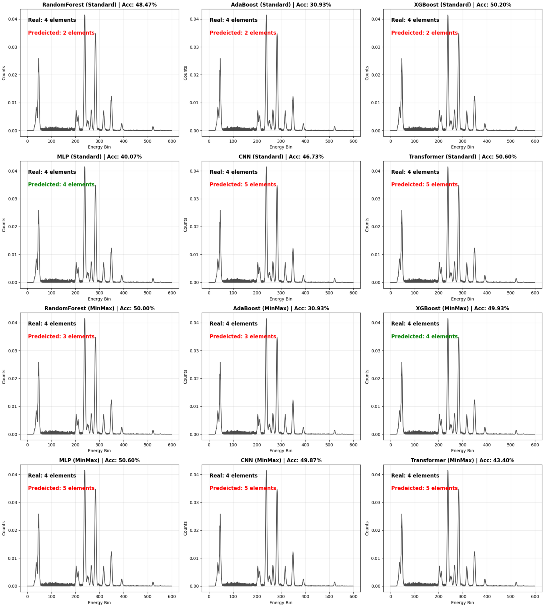
Results Multi-label Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
Standard	XGBoost	0.8691	0.9593	0.8237
MinMax	XGBoost	0.8691	0.9589	0.8243
LogMinMax	XGBoost	0.8691	0.9589	0.8243
MinMax	AdaBoost	0.8205	0.9151	0.7821
LogMinMax	AdaBoost	0.8203	0.9088	0.7830
Standard	AdaBoost	0.8142	0.9049	0.7770
LogMinMax	RandomForest	0.8004	0.9618	0.7314
MinMax	RandomForest	0.8000	0.9600	0.7305
Standard	RandomForest	0.7995	0.9848	0.7292
LogMinMax	CNN	0.7979	0.9719	0.7267
Standard	CNN	0.7910	0.9597	0.7203
MinMax	CNN	0.7808	0.9567	0.7062
MinMax	MLP	0.7567	0.9419	0.6802
LogMinMax	MLP	0.7542	0.9494	0.6751
MinMax	Transformer	0.7338	0.9441	0.6540
Standard	MLP	0.7320	0.9351	0.6415
Standard	Transformer	0.7173	0.8914	0.6331
LogMinMax	Transformer	0.7141	0.9299	0.6295

Results Multi-class Classification



Results: Example Multi-class Classification



Results Multi-class Classification

Preprocessing	Architecture	Accuracy
LogMinMax	MLP	0.5113
Standard	Transformer	0.5060
MinMax	MLP	0.5060
Standard	XGBoost	0.5020
MinMax	RandomForest	0.5000
MinMax	XGBoost	0.4993
LogMinMax	XGBoost	0.4993
MinMax	CNN	0.4987
Standard	RandomForest	0.4847
LogMinMax	CNN	0.4827
Standard	CNN	0.4673
LogMinMax	RandomForest	0.4647
MinMax	Transformer	0.4340
Standard	MLP	0.4007
Standard	AdaBoost	0.3093
MinMax	AdaBoost	0.3093
LogMinMax	AdaBoost	0.3093
LogMinMax	Transformer	0.1760